

MEDICAL CORE · VERIFIED RESEARCH

Mandibular Fracture Reduction Optimization

Convert an ambiguous clinical problem into a testable 3D registration and optimization problem.

EVIDENCE STATE Verified · Completed	PERIOD 2021.12 - 2023.02
PORTFOLIO TIER Medical Core	PRIMARY CAPABILITY 3D Geometry & Registration

Built a surgical-planning simulator and experiment pipeline that optimizes mandibular fragment pose using dental features and fracture surfaces.

Problem and system boundary

01 / PROBLEM DEFINITION

Problem definition

Defined occlusion as a reproducible target for fracture reduction.

PROBLEM

Many geometrically plausible reductions existed; the planning target needed a reproducible constraint grounded in occlusion.

Built a surgical-planning simulator and experiment pipeline that optimizes mandibular fragment pose using dental features and fracture surfaces.

PUBLIC BOUNDARY

Public evidence is the published figures and presentation material; no clinical use or sole first authorship is claimed.

Clinical interpretation, algorithm design, experiment design, and writing were conducted as joint research.

EVIDENCE STATE

Verified · Completed / Surface-distance maps between the pre-fracture mandible and the reduction model (part of Fig. 3 in the published paper).

Owned decisions and implementation

MY ROLE

Jointly led the research pipeline across problem definition, feature and optimization design, simulator, experiments and analysis, and paper, serving as a co-first author.

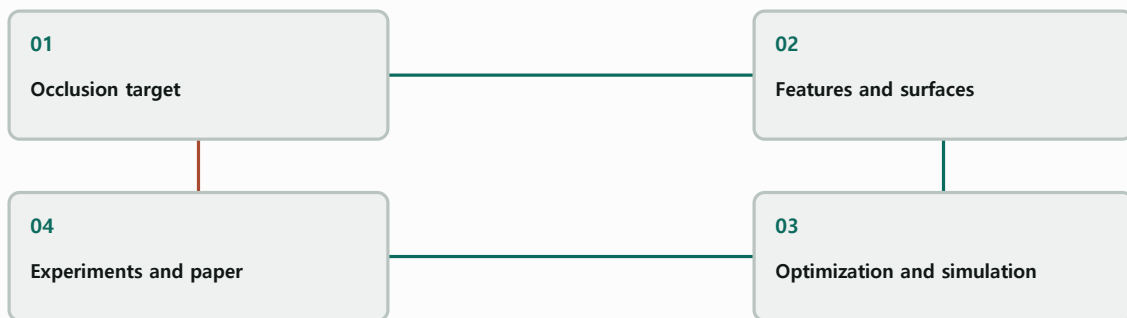
02 / RESEARCH PIPELINE

Research pipeline

Feature design · Optimization · Simulator · Experiments and analysis · Paper

SYSTEM RELATIONSHIP

Occlusion-constrained optimization loop



Explanatory system relationship diagram; it is not photographic or experimental evidence.

IMPLEMENTATION TECHNOLOGIES

Python / Open3D / OpenCV / SciPy / PCA / ICP / CLPSO / 3D Slicer

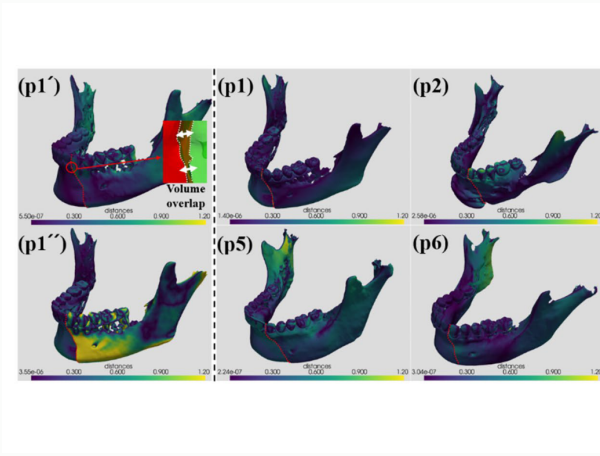
Verification and evidence ledger

03 / PUBLISHED EVIDENCE

Published evidence

Separate presentations, award, quantitative experiments, and the peer-reviewed publication.

IMAGE / PUBLICLY INSPECTABLE



REGISTERED PUBLIC EVIDENCE

PUBLICATION / PUBLICLY INSPECTABLE

mandibular-publication

Public article page used as external research evidence.

[Open public link](#)

IMAGE / PUBLICLY INSPECTABLE

mandibular-fracture-lead-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

mandibular-fracture-gallery-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

mandibular-fracture-gallery-02

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

mandibular-fracture-gallery-03

Approved derivative; caption in portfolio data.

Role, team result, and limits

04 / RESEARCH BOUNDARY

Research boundary

Prepared-dataset results do not establish routine clinical use.

MY ROLE

Jointly led the research pipeline and served as a co-first author.

Jointly led the research pipeline across problem definition, feature and optimization design, simulator, experiments and analysis, and paper, serving as a co-first author.

TEAM RESULT

The joint research team produced international and domestic presentations, an award, and a peer-reviewed publication.

ACCAS 2022, a 2022 domestic conference, a 2023 best-paper award, a 2024 Q1 SCIE peer-reviewed paper, and quantitative experiments provide the evidence.

LIMITATIONS

Public evidence is the published figures and presentation material; no clinical use or sole first authorship is claimed.

Surface-distance maps between the pre-fracture mandible and the reduction model (part of Fig. 3 in the published paper).

COLLABORATION BOUNDARY

Clinical interpretation, algorithm design, experiment design, and writing were conducted as joint research.

Built a surgical-planning simulator and experiment pipeline that optimizes mandibular fragment pose using dental features and fracture surfaces.

Recap and joint-development invitation

Convert an ambiguous clinical problem into a testable 3D registration and optimization problem.

PRIMARY CAPABILITY

3D Geometry & Registration / Medical Navigation & Visualization

REGISTERED PUBLIC EVIDENCE

[Publication](#)

WHAT TO BRING

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

Clinical interpretation, algorithm design, experiment design, and writing were conducted as joint research.

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